Insurance Mathematics and Economics 130 (2026) 103285

Contents lists available at ScienceDirect

Insurance: Mathematics and Economics

journal homepage: www.elsevier.com/locate/ime

# Static marginal expected shortfall: Systemic risk measurement under dependence uncertainty

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ARTICLE INFO

*Keywords:*
Marginal expected shortfall
Risk bounds
Dependence
Systemic risk
Aggregate risk

ABSTRACT

Measuring the contribution of a bank or an insurance company to overall systemic risk is a key concern, particularly in the aftermath of the 2007–2009 financial crisis and the 2020 downturn. In this paper, we derive worst-case and best-case bounds for the marginal expected shortfall (MES)—a key measure of systemic risk contribution—under the assumption that individual firms' risk distributions are known but their dependence structure is unspecified. We further derive tighter MES bounds when partial information on firms' risk exposures, and thus their dependence structure, is available. This partial information is incorporated through two standard factor models: additive and minimum-based background risk models. Additionally, we propose an alternative set of improved MES bounds based on a linear relationship between firm-specific and market-wide risks, consistent with the capital asset pricing model in finance and the weighted insurance pricing model in insurance. Finally, empirical analyses demonstrate the practical relevance of the theoretical bounds for industry practitioners and policymakers.

## 1. Introduction

Understanding and measuring systemic risk, that is, the risk that the failure of a single component could destabilize the entire system, has become a central concern in finance, insurance, and risk management, especially in the aftermath of the 2007–2009 financial crisis and the global economic disruption caused by the COVID-19 pandemic. For example, Canada's Office of the Superintendent of Financial Institutions (OSFI) explicitly highlights the importance of addressing systemic risk arising from 'concentration' in its 2024–2025 Annual Risk Outlook (Office of the Superintendent of Financial Institutions (OSFI), 2024). Furthermore, as noted in Gravelle and Li (2013), a central task in this context is to assess how individual institutions contribute to systemic risk: "*It is a key issue in [prudential] supervision that policymakers need analytical tools to measure the systemic importance of individual institutions.*" The marginal expected shortfall (MES) has emerged as a widely used measure of such systemic risk contribution.

Initially proposed by Acharya et al. (2017)—and long beforehand recognized in actuarial science (Denault, 2001; Panjer and Jing, 2001) as a risk capital allocation (RCA) rule based on the expected shortfall (ES) risk measure—MES has received considerable attention in both research and applications. Speaking more formally, let the random variables (RVs) $X_1, X_2 \dots, X_d$, $d \in \mathbb{N}$ denote the individual risks associated with insurers labeled $\mathcal{N} := \{1, 2, \dots, d\}$, and let the RV $S = X_1 + X_2 + \cdots + X_d$ represent the aggregate risk in the market. Furthermore, we assume that all the aforementioned RVs have cumulative distribution functions (CDFs) with finite means. The ES risk measure of the aggregate risk, for a prudence level $p \in (0, 1)$ and the Value-at-Risk (VaR) risk measure $\mathrm{VaR}_p(S)$, is defined as

$$\mathrm{ES}_p(S) = \mathbb{E}[S|S > \mathrm{VaR}_p(S)],$$

(e.g., Denuit et al., 2006), and the corresponding MES risk measure is defined as

$$\mathrm{MES}_p(X_j, S) = \mathbb{E}[X_j|S > \mathrm{VaR}_p(S)], \ j \in \mathcal{N},$$

as introduced in the systemic risk literature (e.g., Acharya et al., 2017) for dynamic, time-varying frameworks, and in Qin and Zhou (2021) for static, tail-asymptotic, cross-sectional formulations.

The theoretical properties of both the ES and MES functionals have been thoroughly studied. In particular, the former functional has been shown to be coherent (Artzner et al., 1999; Acerbi and Tasche, 2002), comonotonic additive (Kusuoka, 2001), convex (Föllmer and Schied, 2002), elicitable (Fissler and Ziegel, 2016), and uniquely characterized by a set of 'economic' axioms, justifying its role as a leading regulatory

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https://doi.org/10.1016/j.insmatheco.2026.103285
Received 30 April 2025; Received in revised form 10 May 2026; Accepted 16 July 2026
Available online 18 July 2026
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risk measure (Wang and Zitikis, 2021). Furthermore, it is 'distorted' in the sense of Wang (2000) and 'weighted' in the sense of Furman and Zitikis (2008a). Finally, the Solvency III regulatory framework—based on Basel III, which is known as the Fundamental Review of the Trading Book (Basel Committee on Banking Supervision, 2016)—is shifting from the VaR risk measure at the prudence level $p = 0.99$ to the ES risk measure at the prudence level $p = 0.975$ for calculating market risk capital. The latter functional, that is, the MES risk measure, inherits much of the appeal of the ES risk measure on which it is based (e.g., Tsanakas and Barnett, 2003; Furman and Zitikis, 2008b; Dhaene et al., 2012, and references therein).

On a more application-oriented front, a substantial body of literature has derived explicit formulas for the ES and MES functionals under specific assumptions about the joint CDF of the risk RVs $X_1, X_2, \dots, X_d$, $d \in \mathbb{N}$. In particular, Panjer and Jing (2001) provided formulations for the multivariate normal distribution, while Landsman and Valdez (2003) and Dhaene et al. (2008) extended these results to the broader class of elliptical distributions. Furman and Landsman (2005) and Zhou et al. (2018) derived results for a class of multivariate gamma distributions based on additive background risk models, and Furman and Landsman (2010) further extended these findings to a more general class of multivariate Tweedie distributions. The multivariate phase-type and multivariate Pareto distributions were addressed in Cai and Li (2005) and Vernic (2006, 2011), respectively. More recently, Landsman et al. (2016) and Ignatieva and Landsman (2019) derived formulations for risk variables following multivariate log-elliptical and skewed generalized hyperbolic distributions, and Cossette et al. (2018) and Marri and Moutanabbir (2022) considered the case of dependent risks modeled via Archimedean and mixed Bernstein copulas. It should be noted that this overview is far from exhaustive. A considerable body of additional literature has explored ES and MES in diverse modeling frameworks beyond those mentioned here.

Recent actuarial research has increasingly examined systemic risk through MES, ES, and related contribution-based tail risk measures. Li (2022) studied asymptotic results for several forms of MES for dependent risks, while Chen and Liu (2022) analysed systemic ES and MES in a general quantitative risk-management framework. From a statistical-inference perspective, Gribkova et al. (2022) investigated tail conditional allocation, also known as MES, and established its large-sample properties with applications to insurance risk assessment. More recently, Geng et al. (2026) studied systemic ES and MES in a multidimensional renewal risk model with multiple business lines and heterogeneous claims. Beyond MES-specific studies, Dhaene et al. (2022) introduced conditional distortion risk measures and distortion risk contribution measures, and Zhang (2024) further developed distortion risk contribution ratio measures from the perspective of stochastic orders. These studies demonstrate the growing importance of MES-type and contribution-based systemic risk measures in actuarial research.

Specific assumptions on the joint CDF of risks are more of a mathematical nicety than a reflection of reality. In practice, the form of dependence that governs the joint behaviour of the risk RVs $X_1, X_2, \dots, X_d$, $d \in \mathbb{N}$—and hence the distribution of the aggregate risk RV $S$—is rarely known. A plethora of authors have addressed this dependence uncertainty in the context of the aggregate risk $S$. In the case of complete uncertainty, Embrechts et al. (2014) provided a comprehensive discussion and numerical illustrations of such problems in risk management. Bernard et al. (2014) derived a lower bound for $\mathrm{ES}_p(S)$ for homogeneous risks by leveraging properties of complete mixability (Wang and Wang, 2011) and coherent risk measures (Artzner et al., 1999). Jakobsons et al. (2016) further discovered the minimum $\mathrm{ES}_p(S)$ for heterogeneous risks under the assumptions of joint mixability; see Wang and Wang (2016) for the detailed information of joint mixability. Moreover, Wang and Zitikis (2021) proposed the concept of $p$-concentration, which refers to the absence of diversification benefits in portfolio construction, and illustrated the upper bound of $\mathrm{ES}_p(S)$ for any risks is obtained via comonotonicity, as discussed more generally by Dhaene et al. (2002) and Deelstra et al. (2011) in the contexts of finance and insurance. More recently, Bernard et al. (2023) studied sharp lower and upper bounds on the expectation of the product of $d \geq 3$ RVs for some specific marginal distributions. In the case of partial uncertainty, related analyses for $\mathrm{VaR}_p(S)$ and convex risk measures when partial information known from the factor models have been previously studied by Bernard et al. (2017b). Additionally, Bernard et al. (2017a) discovered the improved bounds on VaR of $S$ by imposing its variance constraints.

Measuring the contribution of individual firms to aggregate systemic risk is of central importance in modern prudential supervision, financial regulation, and insurance risk management, as it can support, e.g., the calibration of prudential instruments, capital requirements, and risk-sensitive premium assessment frameworks (Gravelle and Li, 2013). This paper is motivated by a collaboration with Canada's Property and Casualty Insurance Compensation Corporation (PACICC), an industry-funded guarantee scheme—also known as a Policyholder Protection Scheme (PPS)—serving Canada's Property and Casualty insurance sector.

PPSs such as PACICC, State Guaranty Associations (SGAs) in the United States, and National Guarantee Schemes (NGSs) across the European Union charge premiums to their member insurers and provide protection to policyholders by covering claims and unearned premiums in the event of a member insurer's insolvency. At present, virtually all PPSs calculate these premiums primarily based on each member insurer's direct written premiums in covered lines of business—a practice followed by, e.g., PACICC and various SGAs in the United States. However, this approach is limited in that it does not account for an individual member insurer's contribution to aggregate systemic risk. Recognizing this gap, the International Association of Insurance Supervisors has encouraged the development of risk-based premium assessment frameworks for PPSs, as recently discussed in International Association of Insurance Supervisors (IAIS) (2023). Such a shift from direct-written-premium-based premiums to systemic-risk-contribution-based premiums is, however, challenging because PPSs such as PACICC generally do not know the full dependence structure among their member insurers. Thus, even when marginal risk information is available, the joint behaviour of insurers during systemic distress remains uncertain.

Nevertheless, the impact of dependence uncertainty on systemic risk contribution, as measured by MES, remains largely unexplored. Understanding the lower and upper bounds of $\mathrm{MES}_p(X_j, S)$—as well as assessing how close a firm in the market lies to either of these bounds—is critically important for PPSs (also for financial regulators), particularly given the growing prominence of systemic risks, such as cyber threats, climate-related disasters, global pandemics, and geopolitical instability, in today's interconnected risk landscape. Quantifying these bounds provides key insights into worst-case scenarios, supports the design of more resilient financial and insurance systems, and may facilitate fairer and more risk-sensitive participation frameworks for PPSs by aligning member contributions more closely with institution-specific exposure to aggregate systemic distress, thereby potentially mitigating cross-subsidization and related moral-hazard concerns associated with purely premium-volume-based assessments (International Association of Insurance Supervisors, 2013).

In this paper, we address this gap by studying the effect of dependence uncertainty on MES. Our main contributions are as follows. First, we derive sharp lower and upper bounds for $\mathrm{MES}_p(X_j, S)$, together with the dependence structures (copulas) that attain these bounds, under the assumption that only marginal information is available (Section 2). In this setting, the distributions of $X_i$, $i \in \mathcal{N}$, are known, whereas the dependence structure of the risk vector $(X_1, X_2, \dots, X_d)'$, $d \in \mathbb{N}$, is completely unspecified. These bounds characterize the maximal and minimal possible systemic risk contributions of individual firms under extreme dependence scenarios, thereby capturing the full range of feasible MES values consistent with the observed marginals.

Although unconstrained bounds are useful for identifying the full range of MES values with fixed marginal distributions, they may be too wide for practical risk management when some information about

the common drivers of individual risks is available. In many financial and insurance applications, firms are exposed to shared macroeconomic, market-wide, or sector-specific risk factors, while also retaining firm-specific idiosyncratic risk. It is therefore natural to ask how much the feasible range of MES can be reduced once such common-factor information is taken into account. We address this question by extending our analysis to constrained MES bounds that incorporate partial dependence information through factor-model structures (Section 2).

Specifically, we consider models in which the risks $X_i$, $i \in \mathcal{N}$, are represented as

$$X_i = f_i(Y, Z_i), \tag{1.1}$$

where $Y$ denotes a common risk factor and $Z_i$ represents an idiosyncratic risk factor. In this setting, the joint distribution of $(X_i, Y)$ provides partial dependence information linking each individual risk and the common factor, while the full joint dependence structure of $(X_1, X_2, \dots, X_d)'$ remains unspecified. Equivalently, the admissible dependence structures are restricted to those that preserve the marginal distributions of the risks as well as the joint CDFs of $(X_i, Y)$, $i \in \mathcal{N}$. This formulation captures an intermediate situation between complete dependence uncertainty and full specification of the joint distribution.

Factor models are widely used across statistics, economics, finance, and insurance (Bernard et al., 2017b), with applications in asset pricing (Fama and French, 1993), portfolio optimization (Santos et al., 2013), and risk management (Gordy, 2000). In the present context, their role is not to impose a full parametric joint model for all risks, but rather to introduce observable or estimable common-factor (or background factor) exposure that constrains the set of admissible dependence structures. As a result, the dependence-uncertainty spread between the lower and upper MES bounds may be reduced relative to the unconstrained case.

In Section 3, we further examine MES bounds under the assumption of a linear relationship between $\mathbb{E}[X_i \mid S]$ and the aggregate risk $S$, an idea central to both the capital asset pricing model (CAPM) (Levy, 2011) and the weighted insurance pricing model (WIPM) (Furman and Zitikis, 2017b). Under this assumption, we obtain explicit constrained bounds in several settings, including non-negative risks, non-positive risks, and bivariate normal risks. As with factor models, these structural constraints further shrink the uncertainty interval for MES, providing interpretable benchmarks for assessing systemic importance.

Finally, in Section 4, we complement our theoretical results with an empirical analysis of unconstrained and constrained MES bounds using loss data from constituent stocks of the S&P 500 over a specified period. The constrained bounds are computed under the Fama–French five-factor model (FFM) (Fama and French, 2015), illustrating how partial dependence information reduces systemic-risk-contribution uncertainty in practice. The empirical findings confirm the practical relevance of our theoretical framework for industry-funded guarantee schemes and financial regulators. In addition, we propose a systemic risk criticality index (SRCI), which quantifies the relative position of a firm's MES within its own attainable dependence-consistent range. This index provides an interpretable metric for assessing relative systemic criticality: the closer the SRCI is to one, the closer the firm is to its worst-case dependence-consistent systemic-risk contribution. While the upper MES bounds can be used by PPSs and financial supervisors to price participation or allocate capital based on systemic-risk contribution, the SRCI serves as a practical tool for assessing the relative systemic position of firms within the market. During periods of extreme market stress—such as the global financial crisis—an institution's MES may approach its theoretical upper bound, indicating systemic criticality. Conversely, the lower bound can be observed in more diversified market phases, as seen during the pandemic recovery period.

## 2. MES Bounds with fixed marginal distributions

In this section, we derive lower and upper bounds for the MES when the marginal distributions of the risk variables are known, but the dependence structure among them is unspecified. Additionally, it is important to recognize that risk bounds under the assumption of completely uncertain dependence are often of limited practical use. Typically, the gap between the lower and upper bounds is substantial; see Bernard et al. (2017b). Consequently, we also derive tighter and more informative bounds for the MES by incorporating partial dependence information from the structure of factor models.

### 2.1. Preliminaries

Let $(\Omega, \mathcal{F}, \mathbb{P})$ denote a standard atomless probability space, and let $\mathcal{X} := L^0(\Omega, \mathcal{F}, \mathbb{P})$ be the space of all real-valued RVs defined on this space. We denote generic elements of $\mathcal{X}$ by $X$, $Y$, $U$, $X_i$, and $Z_i$, where $i \in \mathcal{N} := \{1, 2, \dots, d\}$ and $d \in \mathbb{N}$. Each $X_i$, $i \in \mathcal{N}$, is interpreted as a risk component, and $U \sim \mathrm{Unif}[0, 1]$ is assumed to be independent of all other RVs unless stated otherwise. Throughout the paper, the notation $X \sim F_X$ indicates that the RV $X$ has CDF $F_X(x) = \mathbb{P}(X \le x)$, $x \in \mathbb{R}$. The quantile function of $X$, for $p \in (0, 1)$, is defined by

$$F_X^{-1}(p) = \inf\{x \in \mathbb{R} : F_X(x) \ge p\}.$$

This quantity is also known as the $\mathrm{VaR}_p(X)$. Unless otherwise specified, all RVs are assumed to have finite expectations.

For the random vector $\mathbf{X} = (X_1, X_2, \dots, X_d)'$ of risk components $X_i \sim F_i$, we denote its joint CDF by

$$F(\mathbf{x}) = \mathbb{P}(\mathbf{X} \le \mathbf{x}),$$

where $\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$, and inequalities are understood componentwise. Further, let $Y \sim H$ be a random variable interpreted as a risk factor. For a given realization $Y = y$, we denote by $\mathbf{X}_{|y}$ and $\mathbf{X}_{|Y}$ the random vectors of conditional risk components $X_{i|y} := X_i|(Y = y) \sim F_{i|y}$ and $X_{i|Y} := X_i|Y \sim F_{i|Y}$, $i \in \mathcal{N}$, respectively. The associated aggregate risks are defined by

$$S := \left(\sum_{i=1}^d X_i\right) \sim F_S, \quad S_{|y} := \left(\sum_{i=1}^d X_{i|y}\right) \sim F_{S_{|y}},$$
$$S_{|Y} := \left(\sum_{i=1}^d X_{i|Y}\right) \sim F_{S_{|Y}}.$$

In what follows, we frequently consider the comonotonic and $j$-opposite monotonic copies of the random vector $\mathbf{X}$, denoted respectively by

$$\mathbf{X}^c = (X_1^c, X_2^c, \dots, X_d^c)' \text{ and } \mathbf{X}^a = (X_1^a, X_2^a, \dots, X_d^a)'.$$

Each of these vectors has coordinates with the same marginal distributions as those of $\mathbf{X}$ but differing dependence structures. Specifically, $\mathbf{X}^c$ exhibits perfect positive dependence (comonotonicity), given by $X_i^c = F_i^{-1}(U)$, $i \in \mathcal{N}$. In contrast, $\mathbf{X}^a$ represents the conceptual $j$-opposite monotone construction—countermonotonicity when $d = 2$—and is constructed as

$$X_j^a = F_j^{-1}(U), \quad X_i^a = F_i^{-1}(1 - U), \quad i \in \mathcal{N} \setminus \{j\}, \quad j \in \mathcal{N}.$$

Analogously to earlier, we write $\mathbf{X}_{|y}^c$ ($\mathbf{X}_{|y}^a$) and $\mathbf{X}_{|Y}^c$ ($\mathbf{X}_{|Y}^a$) for the comonotonic ($j$-opposite monotonic) copies of the conditional random vectors $\mathbf{X}_{|y}$ and $\mathbf{X}_{|Y}$, respectively. The (conditional) aggregate risk RVs in the comonotonic case are constructed as $S^c := F_S^{-1}(U)$, $S_{|y}^c := F_{S_{|y}}^{-1}(U)$, and $S_{|Y}^c := F_{S_{|Y}}^{-1}(U)$, whereas (conditional) aggregate risk RVs in the $j$-opposite monotonic case are constructed as

$$S^a := F_S^{-1}(1 - U), \quad S_{|y}^a := F_{S_{|y}}^{-1}(1 - U), \quad S_{|Y}^a := F_{S_{|Y}}^{-1}(1 - U).$$

Wang and Wang (2016) and Wang and Zitikis (2021) introduced the concepts of joint mixability and risk concentration, respectively, and

established their key properties. We recall these two definitions below and refer the reader to the cited references for further details.

**Definition 1** (Joint mixability). A $d$-tuple of probability distributions $(F_1, F_2, \dots, F_d)$ on $\mathbb{R}$ is said to be jointly mixable if there exist RVs $X_i \sim F_i$, $i \in \mathcal{N}$ such that $\sum_{i=1}^d X_i = c$ a.s., for some constant $c \in \mathbb{R}$, termed the joint center of $(F_1, F_2, \dots, F_d)$. The random vector $(X_1, X_2, \dots, X_d)'$ is called *jointly mixable*.

**Definition 2** (Risk concentration). For $p \in (0,1)$, a random vector $(X_1, X_2, \dots, X_d)'$ is *p-concentrated* if its coordinates share a common tail event of probability $1-p$. A tail event of probability $1-p$ is called a $p$-tail event.

Before presenting the main results of this section, we introduce the left expected shortfall (LES) as a measure of left-tail risk.

**Definition 3** (LES). The LES risk measure of a RV $X \in \mathcal{X}$ at the prudence level $p \in (0,1)$, denoted by $\mathrm{LES}_p(X)$, is

$$\mathrm{LES}_p(X) = \mathbb{E}[X | X \le \mathrm{VaR}_p(X)].$$

### 2.2. Risk bounds

Let $G_i(x_i, y) := \mathbb{P}(X_i \le x_i, Y \le y)$ be the joint CDF of the RVs $X_i$ and $Y$, $i \in \mathcal{N}$. Specifically, $G_i$ contains the partial dependence information obtained from the factor models. Let $m_j(p)$ ($M_j(p)$) and $m_j^f(p)$ ($M_j^f(p)$) denote the unconstrained and constrained lower (upper) bounds of the functional $\mathrm{MES}_p(X_j, S)$, $j \in \mathcal{N}$, respectively:

$$m_j(p) = \inf\Big\{\mathrm{MES}_p(X_j,S) : \ S = \sum_{i=1}^d X_i, \ X_i \sim F_i\Big\},$$

$$M_j(p) = \sup\Big\{\mathrm{MES}_p(X_j,S) : \ S = \sum_{i=1}^d X_i, \ X_i \sim F_i\Big\},$$

$$m_j^f(p) = \inf\Big\{\mathrm{MES}_p(X_j,S) : \ S = \sum_{i=1}^d X_i,$$
$$X_i = f_i(Y,Z_i) \sim F_i, \ (X_i,Y) \sim G_i, Y \sim H, Y \perp Z_i\Big\},$$

$$M_j^f(p) = \sup\Big\{\mathrm{MES}_p(X_j,S) : \ S = \sum_{i=1}^d X_i,$$
$$X_i = f_i(Y,Z_i) \sim F_i, \ (X_i,Y) \sim G_i, Y \sim H, Y \perp Z_i\Big\}.$$

**Theorem 1** (Unconstrained and constrained risk bounds). *Let $X_i = f_i(Y, Z_i) \sim F_i$, where $i \in \mathcal{N}$, $Y \sim H$, and $Z_i$ are independent of $Y$, and let $S = \sum_{i=1}^d X_i$. Then, for $p \in (0,1)$ and $j \in \mathcal{N}$, we have*

$$\begin{aligned}\mathrm{MES}_p(X_j^a, S^a) \le m_j(p) \le \mathrm{MES}_p(X_{j|Y}^a, S_{|Y}^a) \le m_j^f(p) \le \mathrm{MES}_p(X_j, S)\\ \le M_j^f(p) := \mathrm{MES}_p(X_{j|Y}^c, S_{|Y}^c)\\ \le M_j(p) := \mathrm{MES}_p(X_j^c, S^c).\end{aligned} \tag{2.1}$$

The proofs of this and all subsequent theorems and propositions are relegated to the Appendix. The (improved) upper bound of $\mathrm{MES}_p(X_j, S)$ is always sharp due to (conditionally) comonotonicity. The comonotonic random vector $\mathbf{X}^c$ yields sharp upper bounds for various risk measures, including law-invariant convex risk measures; see, e.g., Meilijson and Nádas (1979), Puccetti and Rüschendorf (2012) and Bernard et al. (2017b).

**Proposition 1.** *Let $X_i \sim F_i$, where $i \in \mathcal{N}$, and $X_i$ have continuous CDFs, and let $S = \sum_{i=1}^d X_i$. Then, for $p \in (0,1)$ and $j \in \mathcal{N}$, the unconstrained risk bounds in (2.1) reduce to*

$$\mathrm{LES}_{1-p}(X_j) \le m_j(p) \le \mathrm{MES}_p(X_j, S) \le M_j(p) = \mathrm{ES}_p(X_j). \tag{2.2}$$

*Furthermore, the upper bound is obtained when the random vector $(X_1, X_2, \dots, X_d)'$ is p-concentrated.*

The upper bound $\mathrm{ES}_p(X_j)$ is well-known due to a property of RCAs known as 'no-undercut'. That is, no individual risk component $X_j$, $j \in \mathcal{N}$ is allocated more risk capital than it would be allocated if considered alone (Furman and Zitikis, 2008b). In this case, $M_j(p)$ is independent of the dimension of the random vector and can be computed explicitly from the marginal distribution of the risk component $X_j$ (the worst-case MES of $X_j$ at prudence level $p$ coincides with ES of the marginal risk). The lower bound $m_j(p)$ in Inequality (2.2) implies that the minimum systemic risk contribution of an individual risk component $X_j$ cannot be smaller than its conditional expected value, given that the loss does not exceed its VaR at the prudence level $1-p$.

Fig. 1 presents the unconstrained and constrained risk bounds of $\mathrm{MES}_p(X_j, S)$ as a function of the prudence level $p$ for the case $X_i = \exp\left(b_i Y + \sqrt{1-b_i^2} Z_i\right) \sim \mathrm{Lognorm}(0,1)$, where $i = 1,2$, $b_1 = 0.1$, $b_2 = 0.9$, $Y \overset{d}{=} Z_i \sim N(0,1)$. The vertical dashed line illustrates the full range of possible MES values, from the minimum (LES) to the maximum (ES), while the solid one shows its improved range of values. As shown,

$$\begin{aligned}\lim_{p\to 0}\mathrm{MES}_p(X_j,S) &= \lim_{p\to 0}\mathrm{LES}_{1-p}(X_j) = \lim_{p\to 0}\mathrm{ES}_p(X_j)\\ &= \lim_{p\to 0} m_j^f(p) = \lim_{p\to 0} M_j^f(p) = \mu_j.\end{aligned}$$

In contrast to the (improved) upper bound of $\mathrm{MES}_p(X_j, S)$ in Theorem 1, which is attained under (conditional) comonotonicity, the (improved) lower bound $m_j(p)$ (or $m_j^f(p)$) is not necessarily sharp. In what follows, we examine the conditions under which this lower bound becomes sharp, with particular attention to the role of the (conditional) marginal distributions.

### 2.3. On the sharpness of the lower bounds

In this subsection, we first identify the conditions under which $m_j(p)$ and $m_j^f(p)$ in Theorem 1 are attained and subsequently demonstrate sharpness through examples and counterexamples.

**Corollary 1** (Sharpness of the lower bounds). *Let $X_i = f_i(Y, Z_i) \sim F_i$, $i \in \mathcal{N}$, be $d$ risks, and $S = \sum_{i=1}^d X_i$ denote the aggregate risk. The lower bound of $m_j(p)$ in Theorem 1 is attained if and only if $X_j$ and $S$ are countermonotonic, while the lower bound of $m_j^f(p)$ is attained when $X_{j|Y}$ and $S_{|Y}$ are countermonotonic.*

The proof of Corollary 1 follows from Theorem 1 and is therefore omitted.

**Remark 1.** The (conditional) countermonotonicity condition in Corollary 1 may appear to hold universally since it involves only two RVs, $X_j$ and $S$. However, as $S = X_j + \sum_{i\in\mathcal{N}\setminus\{j\}} X_i$ represents an aggregate risk, the validity of this condition depends on the properties of the (conditional) marginal distributions $F_i$ ($F_{i|Y}$).

The following examples illustrate the point of Remark 1.

#### 2.3.1. The unconstrained lower bound

To demonstrate that $m_j(p) > \mathrm{MES}_p(X_j^a, S^a)$ in (2.1), we provide several illustrative examples.

**Example 1.** Let $X_1 \sim F_1$ and $X_2 \sim F_2$ denote two discrete RVs having respective mass points $\{-2,-1,0,1,2\}$ and $\{-1,1,3,5,7\}$ with equal probability and $p = .6$. Under the countermonotonic dependence between $X_1$ and $X_2$, $0 = \mathrm{MES}_p(X_2^a, S^a) < m_2(p) = M_2(p) = 6$ since $S$ and $X_2$ are comonotonic.

An example with normal marginal distributions is provided to further illustrate this result.

**Example 2.** Let $X_i \sim N(\mu_i, \sigma_i^2)$, $i \in \mathcal{N}$, such that $\sum_{i\in\mathcal{N}\setminus\{j\}} \sigma_i < \sigma_j$, and $S = \sum_{i=1}^d X_i$. Then the lower bound of $m_j(p)$ is

$$\mathrm{LES}_{1-p}(X_j) = \frac{\mu_j - \sigma_j \phi\big(\Phi^{-1}(1-p)\big)}{1-p}.$$

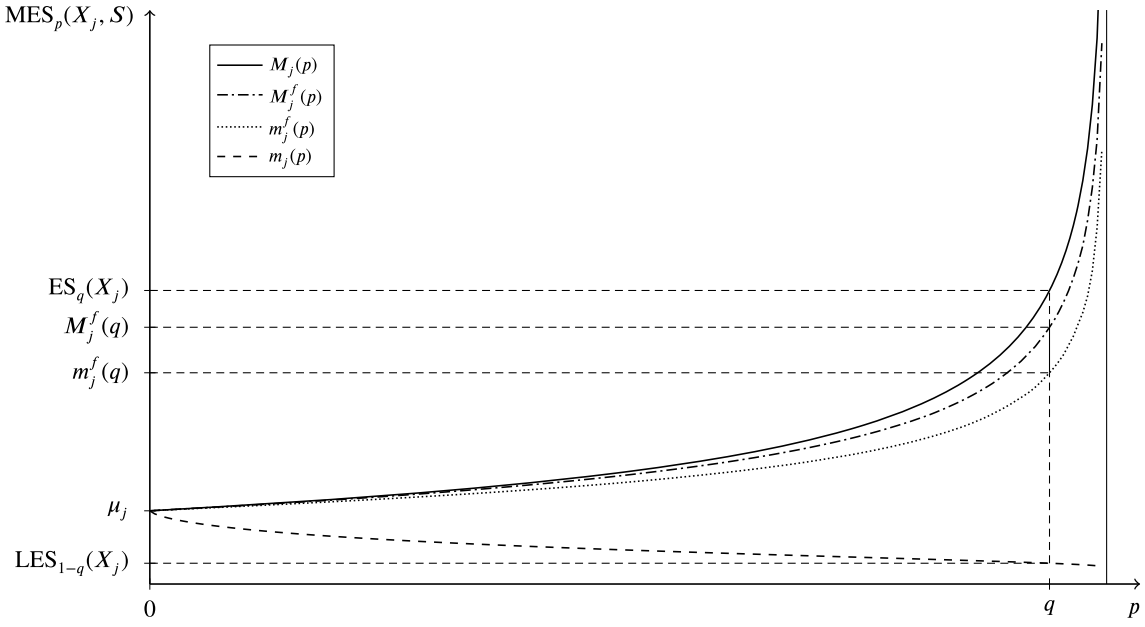



**Fig. 1.** Risk bounds of $\mathrm{MES}_p(X_j, S)$, where $X_i \sim \mathrm{Lognorm}(0,1)$ and $S = \sum_{i=1}^2 X_i$, as functions of the prudence level $p$. Under the assumptions of factor models, $X_1 = \exp(b_1 Y + \sqrt{1-b_1^2} Z_1)$ and $X_2 = \exp(b_2 Y + \sqrt{1-b_2^2} Z_2)$, where $b_1 = 0.1$, $b_2 = 0.9$, and $Y \overset{d}{=} Z_i \sim N(0,1)$.

However, $m_j(p) = M_j(p) > \mathrm{LES}_{1-p}(X_j)$ is obtained when $X_j = \mu_j + \sigma_j \Phi^{-1}(U)$ and $X_i = \mu_i + \sigma_i \Phi^{-1}(1-U)$ for $i \in \mathcal{N}/\{j\}$. In this case, $S = \sum_{i=1}^d \mu_i + \left(\sigma_j - \sum_{i \in \mathcal{N}\setminus\{j\}} \sigma_i\right)\Phi^{-1}(U)$ is comonotonic to $X_j$.

We next provide propositions and examples illustrating when the lower bound of $m_j(p)$ is sharp.

**Proposition 2.** *Let $X_i \sim F_i$ for $i \in \mathcal{N}$, be $d$ risks such that the tuple of distributions $(F_i, F_j)$ is jointly mixable for some $i \neq j$. Then, there exists a random vector $(X_1, X_2, \dots, X_d)$ such that $X_j$ and $S = \sum_{i=1}^d X_i$ are countermonotonic. In this case, the lower bound of $m_j(p)$ in Theorem 1 is obtained. Furthermore, it holds that $X_j = F_j^{-1}(U)$ and $X_i = F_i^{-1}(1-U)$ for all $i \in \mathcal{N}/\{j\}$.*

We omit the proof of Proposition 2 since it follows directly. Under this construction, we have

$$S = c + \sum_{k=1\&k\neq i,j}^{d} F_k^{-1}(1-U) = F_S^{-1}(1-U), \tag{2.3}$$

where $c \in \mathbb{R}$ is the joint center of $(F_i, F_j)$. Specifically, if $F_i$ in Proposition 2 are identical, symmetric and continuous, $m_j(p) = \mathrm{LES}_{1-p}(X_j)$ is obtained by a random vector $(X_1, X_2, \dots, X_d)$ such that $X_j$ and $S$ are countermonotonic. Furthermore, $X_j = F^{-1}(1-U)$ and $X_i = F^{-1}(U)$ for $i \in \mathcal{N}/\{j\}$. In this case, $c = 2\mu$ in Equation (2.3), where $\mu = \mathbb{E}[X_i]$.

**Example 3.** Let $U_i \sim \mathrm{Unif}[0,1]$, $i \in \mathcal{N}$, be $d \geq 3$ risks, and $S = \sum_{i=1}^d U_i$. Hence,

$$m_j(p) = \mathrm{LES}_{1-p}(U_j) = \frac{\int_0^{1-p} u du}{1-p} = \frac{1-p}{2}$$

is obtained when $U_j = U$ and $U_i = 1-U$ for $i \in \mathcal{N}/\{j\}$.

### 2.3.2. The constrained lower bound

To illustrate that $m_j^f(p) > \mathrm{MES}_p(X_{j|Y}^a, S_{|Y}^a)$ and $m_j^f(p) = \mathrm{MES}_p(X_{j|Y}^a, S_{|Y}^a)$ in (2.1), we consider the following examples.

**Example 4.**

(1) Let $X_i = Y + Z_i$, $i = 1, 2$ be two risks such that $Z_i \sim \mathrm{Unif}[0,1]$ are independent of $Y \sim \mathrm{Unif}[0,1]$. In this case,

$$\begin{aligned} m_j^f(p) &= \mathbb{E}[Y + Z_j | 2Y + U + 1 - U > \mathrm{VaR}_p(2Y + U + 1 - U)] \\ &= \mathbb{E}[Y + Z_j | Y > \mathrm{VaR}_p(Y)] = \mathbb{E}[Y | Y > \mathrm{VaR}_p(Y)] + \mathbb{E}Z_j \\ &= \frac{1+p}{2} + \frac{1}{2} > \mathrm{MES}_p(X_{j|Y}^a, S_{|Y}^a). \end{aligned}$$

The lower bound of $m_j^f(p)$ is not obtained since $X_j$ and $S$ cannot be conditionally countermonotonic.

(2) Let $\boldsymbol{X} = (X_1, X_2, X_3)' = Y(-1,1,1)' + (Z_1, Z_2, Z_3)'$ be a three-dimensional random vector such that $Y \sim \mathrm{N}(0,1)$ and $Z_i \sim \mathrm{N}(0,1)$ are independent of $Y$. Hence, $X_i \sim \mathrm{N}(0,2)$. The lower bound of $\mathrm{MES}_p(X_2, S)$ is

$$m_2^f(p) = \mathbb{E}[Y + Z_2 | Y - Z_2 > \mathrm{VaR}_p(Y - Z_2)] = 0$$

since $Y + Z_2$ and $Y - Z_2$ are independent, and it is obtained when $Z_2 = \Phi^{-1}(U)$ and $Z_1 = Z_3 = \Phi^{-1}(1-U)$. In this case, $X_2$ and $S$ are conditionally countermonotonic.

### 2.4. Examples

We now provide examples to illustrate how partial dependence information reduces the feasible range of MES. To capture partial dependence, we employ two widely used factor models: the additive background risk model (ABRM) and the minimum-based background risk model (MBRM). For ABRM and $i \in \mathcal{N}$, the functional form is $f_i(Y, Z_i) = \mu_i + b_i Y + \sigma_i Z_i$ with $\sigma_i > 0$, while $f_i(Y, Z_i) = \min(Y, Z_i)$ for MBRM. We refer the reader to Fama and French (2004) and Furman et al. (2018) for ABRM, and Asimit et al. (2010) and Pai and Ravishanker (2020) for MBRM.

To quantify the improvement in the dependence-uncertainty spread achieved by introducing a factor model, we define the following measure, denoted by $\delta_{j,p}$:

$$\delta_{j,p} = 1 - \frac{M_j^f(p) - m_j^f(p)}{M_j(p) - m_j(p)},$$

where we conventionally set $\delta_{j,p} = 1$ whenever $M_j(p) = m_j(p)$; see a similar definition for general risk measures in Bernard et al. (2017b). Here,

$M_j(p) - m_j(p)$ and $M_j^f(p) - m_j^f(p)$ are referred to as the dependence-uncertainty spreads of $\mathrm{MES}_p(X_j, S)$ in the unconstrained and constrained setting, respectively.

We now consider an example in which each risk component $X_i$ follows a standard normal distribution. This example considers a classical Gaussian one-factor additive background risk model, which is closely related to dependence structures widely used in finance, actuarial science, and risk management.

**Example 5** (Additive background risk models)**.** Let $X_i = b_i Y + \sqrt{1-b_i^2}Z_i$, $i = 1, 2$, be two standard normally distributed risks such that $b_i \in (-1, 1)$, $Z_i \sim N(0, 1)$ are independent of $Y$, and $(X_i, Y) \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with the mean vector $\boldsymbol{\mu} = (0, 0)$ and the covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & b_i \\ b_i & 1 \end{bmatrix}$. Moreover, we denote the aggregate risk by $S = \sum_{i=1}^2 b_i Y + \sqrt{1-b_i^2}Z_i$. For the unconstrained bounds of $\mathrm{MES}_p(X_j, S)$, we obtain that

$$M_j(p) = \frac{\phi\big(\Phi^{-1}(p)\big)}{1-p}, \text{ and } m_j(p) = 0.$$

Under the ABRM setting, the constrained bounds of $\mathrm{MES}_p(X_j, S)$ are

$$M_j^f(p) = \frac{b_j(b_1+b_2) + \sqrt{1-b_j^2}\sum_{i=1}^2\sqrt{1-b_i^2}}{\sqrt{2\Big(1 + b_1 b_2 + \sqrt{(1-b_1^2)(1-b_2^2)}\Big)}} \frac{\phi\big(\Phi^{-1}(p)\big)}{1-p},$$

and

$$m_j^f(p) = \frac{b_j(b_1+b_2) + \sqrt{1-b_j^2}\Big(\sqrt{1-b_j^2} - \sqrt{1-b_i^2}\Big)}{\sqrt{2\Big(1 + b_1 b_2 - \sqrt{(1-b_1^2)(1-b_2^2)}\Big)}} \frac{\phi\big(\Phi^{-1}(p)\big)}{1-p},$$

where $i, j = 1, 2$, and $i \neq j$. Hence, the degree of improvement we obtain by adding the ABRM-type information is

$$\delta_{j,p} = 1 - \frac{b_j(b_1+b_2) + \sqrt{1-b_j^2}\sum_{i=1}^2\sqrt{1-b_i^2}}{\sqrt{2\Big(1 + b_1 b_2 + \sqrt{(1-b_1^2)(1-b_2^2)}\Big)}} + \frac{b_j(b_1+b_2) + \sqrt{1-b_j^2}\Big(\sqrt{1-b_j^2} - \sqrt{1-b_i^2}\Big)}{\sqrt{2\Big(1 + b_1 b_2 - \sqrt{(1-b_1^2)(1-b_2^2)}\Big)}}.$$

Interestingly, under the ABRM setting with standard normal marginals, the measure of improvement $\delta_{j,p}$ is independent of the prudence level $p$. The next example illustrates the bounds when the individual risks $X_i$ for $i = 1, 2$ are exponentially distributed and $f_i$ is MBRM.

**Example 6** (Minimum-based background risk model)**.** Let $X_i = \min(Y, Z_i)$, $i = 1, 2$, be two risks such that $Z_i \sim \mathrm{Expo}(\lambda)$ follow the exponential distribution with rate parameter $\lambda$, and $Y \sim \mathrm{Expo}(\lambda_0)$ is independent of $Z_i$. Under this setting, it follows that $X_i \sim \mathrm{Expo}(\lambda_0 + \lambda)$. For the unconstrained bounds of $\mathrm{MES}_p(X_1, S)$, we obtain

$$M_1(p) = \frac{1 - \ln(1-p)}{\lambda_0 + \lambda}, \; m_1(p) = -\frac{1}{\lambda_0 + \lambda}\mathbb{E}\big[\ln(1-U) \,|\, -\ln(U - U^2) > u_p\big],$$

where $u_p = \mathrm{VaR}_p(-\ln(U - U^2))$. Under the MBRM setting, the constrained bounds for $\mathrm{MES}_p(X_1, S)$ are given by

$$M_1^f(p) = \frac{1 - \ln(1-p)}{\lambda_0 + \lambda},$$

and

$$m_1^f(p) = \mathbb{E}\left[\min\left(\frac{f(U_1)}{\lambda_0}, \frac{f(U)}{\lambda}\right) \middle| \min\left(\frac{f(U_1)}{\lambda_0}, \frac{f(U)}{\lambda}\right) + \min\left(\frac{f(U_1)}{\lambda_0}, -\frac{\ln(U)}{\lambda}\right) > v_p\right],$$

where $U \perp U_1 \sim \mathrm{Unif}[0, 1]$, $f(x) = -\ln(1-x)$, and

$$v_p = \mathrm{VaR}_p\left(\min\left(\frac{f(U_1)}{\lambda_0}, \frac{f(U)}{\lambda}\right) + \min\left(\frac{f(U_1)}{\lambda_0}, -\frac{\ln(U)}{\lambda}\right)\right).$$

The upper bound $M_1^f(p)$ is attained when the RVs $Z_i$ are comonotonic, whereas the lower bound $m_1^f(p)$ corresponds to the $j$-opposite monotonic case. Notably, under the MBRM framework, the unconstrained and constrained upper bounds coincide. While explicit closed-form solutions for $m_1(p)$ and $m_1^f(p)$ are unavailable, they can be approximated via simple numerical methods by simulating $U$ and $U_1$.

Example 6 demonstrates that MBRMs can reduce the spread of dependence uncertainty by incorporating additional information about the dependence structure. However, the marginal distribution parameters significantly influence the degree of improvement, and for certain parameter configurations, the effect may even vanish.

## 3. Alternative constrained risk bounds

In this section, we derive a different form of constrained risk bounds compared to those for factor models discussed previously. Specifically, we assume that the conditional expectation of each individual RV has a common linear relationship with the aggregate sum of all risks. For $\alpha, \beta \in \mathbb{R}$, we formulate the corresponding problems as

$$m_j^l(p) = \inf\left\{\mathrm{MES}_p(X_j, S) : S = \sum_{i=1}^d X_i, X_i \sim F_i, \mathbb{E}[X_i|S] = \alpha + \beta S\right\},$$

$$M_j^l(p) = \sup\left\{\mathrm{MES}_p(X_j, S) : S = \sum_{i=1}^d X_i, X_i \sim F_i, \mathbb{E}[X_i|S] = \alpha + \beta S\right\}.$$

We first recall the following theorem regarding the WIPM from Furman and Zitikis (2008b, 2017a) and references therein.

**Theorem 2** (WIPM)**.** *Let $X_i \sim F_i$, $i \in \mathcal{N}$, be $d$ risks and $S = \sum_{i=1}^d X_i$. Let $\Pi_w(X_j, S) = \frac{\mathbb{E}[X_j w(S)]}{\mathbb{E}[w(S)]}$ and $\pi_w(S) = \Pi_w(S, S)$, where $w(\cdot)$ is a weight function. If there exist constants $\alpha, \beta \in \mathbb{R}$ such that $\mathbb{E}[X_i|S] = \alpha + \beta S$ for all $i$, then*

$$\Pi_w(X_j, S) = \mathbb{E}[X_j] + \beta\big(\pi_w(S) - \mathbb{E}[S]\big)$$

*holds for every non-decreasing weight function $w(\cdot)$ for which all quantities under consideration are well-defined and finite.*

**Remark 2.** The weighted RCA rule $\Pi_w(X_j, S)$ is a versatile risk measure that enables the study of different risk problems through appropriate choices of $w$. For example, by choosing $w(s) = \mathbb{1}_{s>t}$, where $t \in \mathbb{R}$, $\Pi_w(X_j, S)$ reduces to the systemic risk contribution measure $\mathrm{MES}_p(X_j, S)$. We refer to Furman and Zitikis (2008b) for further details on different weighted allocation rules corresponding to various choices of $w$.

**Proposition 3.** *Let $(X, S) \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where the mean vector $\boldsymbol{\mu} = (\mu_X, \mu_S)$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_S \\ \rho\sigma_X\sigma_S & \sigma_S^2 \end{bmatrix}$. The constrained lower and upper bounds of $\mathrm{MES}_p(X, S)$ are*

$$\mu_X - \frac{\sigma_X}{\sigma_S}\big(\mathrm{ES}_p(S) - \mu_S\big) \leq \mathrm{MES}_p(X, S) \leq \mu_X + \frac{\sigma_X}{\sigma_S}\big(\mathrm{ES}_p(S) - \mu_S\big). \quad (3.1)$$

**Remark 3.** Proposition 3 can be further extended to the case where $(X, S)$ follows a bivariate elliptical distribution. In this generalization, the intercept $\alpha$ and slope $\beta$ retain the same forms as in the normal case (Fang, 2018), although $\sigma_S^2$, $\sigma_X^2$ and $\rho\sigma_X\sigma_S$ may not represent the actual variances and covariance.

**Theorem 3.** *Let $X_i \sim F_i$, $i \in \mathcal{N}$, be $d$ non-negative (or non-positive) and continuous risks with $\mu_i = \mathbb{E}[X_i]$, and $S = \sum_{i=1}^d X_i$. Suppose there exist constants $\alpha, \beta \in \mathbb{R}$ such that $\mathbb{E}[X_i|S] = \alpha + \beta S$. Then, the constrained lower and upper bounds for $\mathrm{MES}_p(X_j, S)$ are given by*

$$\mu_j \le m_j^I(p) \le \mathrm{MES}_p(X_j, S) \le r_j \sum_{i=1}^d \mathrm{ES}_p\big(X_i\big) = M_j^I(p), \tag{3.2}$$

*where $r_j = \frac{\mu_j}{\sum_{i=1}^d \mu_i}$. Furthermore, the lower bound $\mu_j$ is sharp if there exists $Y_j \overset{d}{=} X_i$ such that $(Y_1, Y_2, \dots, Y_d)$ is jointly mixable. The upper bound $r_j \sum_{i=1}^d \mathrm{ES}_p\big(X_i\big)$ is obtained when $X_i = F_i^{-1}(U)$ with $U \sim \mathrm{Unif}[0,1]$.*

**Remark 4.** To achieve the sharp lower bound in (3.2), joint mixability of $(F_1, F_2, \dots, F_d)$ must be established, which in general is a nontrivial problem. Necessary and sufficient conditions for joint mixability for various classes of distributions are provided in Wang and Wang (2016); see also the initial concept introduced by Wang and Wang (2011). Therefore, the sharpness of the lower bound is affected by the nature of the distribution functions. The upper bound, however, is always sharp and corresponds to the comonotonic case.

The following example illustrates a case where the linear conditional expectation assumption is satisfied.

**Example 7.** Let $X_i \sim \mathrm{Unif}[0,1]$, $i = 1,2,3$, where $\mu_i = \frac{1}{2}$, and $S = X_1 + X_2 + X_3$ denote the aggregate risk. The unconstrained bounds for $\mathrm{MES}_p(X_j, S)$ are

$$M_j(p) = \frac{1}{2}(1+p) \text{ and } m_j(p) = \frac{1}{2}(1-p).$$

Assuming $\mathbb{E}[X_i|S] = \frac{1}{3}S$, the constrained bounds for $\mathrm{MES}_p(X_j, S)$ are

$$M_j^I(p) = \frac{1}{2}(1+p) \text{ and } m_j^I(p) = \frac{1}{2}.$$

Furthermore, the lower bound $\frac{1}{2}$ is obtained when

$$Y_1 = U \sim \mathrm{Unif}[0,1];$$

$$Y_2 = \begin{cases} -2U + 1, & \text{if } 0 \le U \le \frac{1}{2}; \\ -2U + 2, & \text{if } \frac{1}{2} < U \le 1; \end{cases}$$

$$Y_3 = \begin{cases} U + \frac{1}{2}, & \text{if } 0 \le U \le \frac{1}{2}; \\ U - \frac{1}{2}, & \text{if } \frac{1}{2} < U \le 1. \end{cases}$$

Note that $(Y_1, Y_2, Y_3)'$ is jointly mixable with joint center $c = \frac{3}{2}$, i.e., $\sum_{i=1}^3 Y_i = \frac{3}{2}$ almost surely; see Rüschendorf and Uckelmann (2002) for the concept of the mixing copula. The upper bound $\frac{1}{2}(1+p)$ is obtained when $X_i = U$ for each $i$.

The degree of improvement is $\delta_{j,p} = \frac{1}{2}$, which is independent of $p$. Importantly, while an improvement in the lower bound exists:

$$m_j(p) = \frac{1}{2}(1-p) \le \frac{1}{2} = m_j^I(p),$$

the upper bound remains unchanged.

## 4. An empirical study with real data

In this section, we conduct empirical analyses on risk bounds under full dependence uncertainty, as well as on improved risk bounds within the framework of the FFM, which is an ABRM as in Section 2.4.

Let $Z_i$ be idiosyncratic components. We consider a vector of risk components $\boldsymbol{X} = \big(X_1, X_2, \dots, X_d\big)'$ with

$$X_i = X_r + \mu_i + \boldsymbol{b_i'Y} + Z_i, \tag{4.1}$$

where $X_r$ denotes the risk-free loss (i.e., the negative of the risk-free return), $\mu_i$ represents abnormal losses, $\boldsymbol{Y} = (Y_1, Y_2, Y_3, Y_4, Y_5)'$ contains the Fama-French five factors, and $\boldsymbol{b_i} = (b_{i1}, b_{i2}, b_{i3}, b_{i4}, b_{i5})'$ are the corresponding factor exposures; $i \in \mathcal{N}$. Although the standard FFM is often used with strong assumptions on the idiosyncratic terms, our purpose here is not to impose a specific joint distribution on $Z_1, Z_2, \dots, Z_d$. Instead, we use the FFM only to identify the common component of each loss and then study the remaining dependence uncertainty induced by the idiosyncratic components.

In the remainder of this section, the dependence constraint used to obtain the improved bounds is the Fama–French five-factor structure in Eq. (4.1). Specifically, the random vector $\boldsymbol{X}$ preserves partial dependence information through the joint CDFs of $(X_i, X_r, Y_1, Y_2, Y_3, Y_4, Y_5)'$, as implied by the fitted FFM. Hence, the dependence between each individual loss and the common risk factors is fixed, while the remaining uncertainty in the joint dependence structure of $X_1, X_2, \dots, X_d$ is represented through the admissible dependence structures of the idiosyncratic components $Z_1, Z_2, \dots, Z_d$. The improved bounds $m_j^f(p)$ and $M_j^f(p)$ are therefore computed by optimizing over these admissible dependence structures.

For the empirical analysis, we use daily losses of stocks that were continuously included in the S&P 500 index during the sample period to represent the random losses $X_i$ and the aggregate loss $S = \sum_{i=1}^d X_i$. In these experiments, the individual risk is represented by the loss of a selected constituent $X_j$, while the market risk corresponds to the sum of losses of all constituents in the index ($S$). Historical returns and constituent data for the S&P 500 are obtained from the CRSP dataset in Wharton Research Data Services. Losses are computed as the negative of returns. The daily Fama-French five factors are collected from the Kenneth R. French Data Library[1] The abnormal losses $\mu_i$ and factor exposures $\boldsymbol{b_i}$ in the FFM are estimated using generalized linear models (GLMs) via the Python package `statsmodels`, fitted on daily losses and factor data. For simplicity, we do not discuss model fit in detail, as it is not the primary focus of this paper. Nevertheless, we note that the GLMs exhibit strong explanatory performance.

Throughout the experiments, empirical daily losses, factors, and risk-free losses are treated as sampled values for the loss vector $\boldsymbol{X}$, factor vector $\boldsymbol{Y}$, and $X_r$, respectively. We then calculate the unconstrained risk bounds for MES, $M_j(p)$ and $m_j(p)$, and the constrained bounds, $M_j^f(p)$ and $m_j^f(p)$ (Theorem 1). For the unconstrained upper (resp., lower) bound, we apply the rearrangement algorithm (see, e.g., Puccetti and Rüschendorf, 2012; Bernard et al., 2026) to the empirical samples of $X_j$ and $S$ so that they are comonotonic (resp., countermonotonic). That is, for the upper (resp., lower) bound, we sort $X_j$ increasingly and sort $S$ increasingly (resp., decreasingly).

For the constrained bounds, we apply the same idea to the idiosyncratic components. Specifically, after estimating $\mu_i$ and $\boldsymbol{b}_i$, we compute the fitted component

$$K_i = X_r + \mu_i + \boldsymbol{b}_i'\boldsymbol{Y}$$

and the residual component

$$Z_i = X_i - K_i.$$

The common components $K_i$ are then kept fixed, while the residuals $Z_i$ are rearranged. The upper constrained bound is obtained by arranging $Z_j$ and $\sum_{i=1}^d Z_i$ comonotonically, whereas the lower constrained bound is obtained by arranging them in the opposite order. This procedure preserves the FFM-implied dependence between the individual losses and the common factors, while varying the remaining unknown dependence among the idiosyncratic components. For comparison, we also compute the empirical MES directly from the sampled values of $\boldsymbol{X}$. The prudence level $p$ is taken from linearly spaced values in $[0, 0.99]$. Note that when $p = 0$, all five quantities reduce to the sample mean of $X_j$.

[1] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

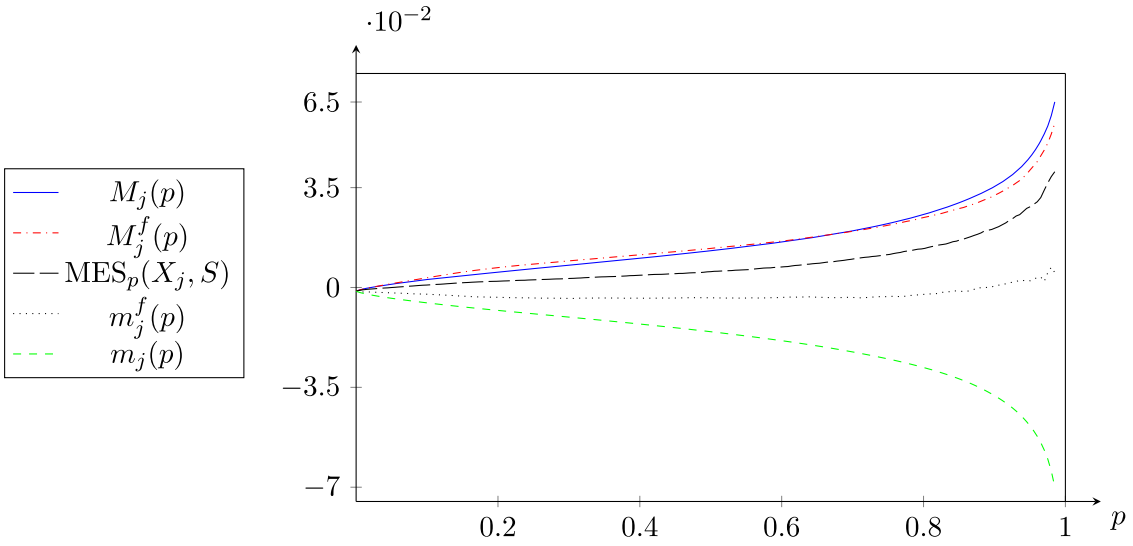



**Fig. 2.** Unconstrained and constrained risk bounds, together with the empirical values of $\mathrm{MES}_p(X_j, S)$, as functions of the prudence level $p$. $X_i$ denotes the daily loss of the $i$-th S&P 500 constituent that remained continuously in the index from 1 January 2005 to 31 December 2024 ($i = 1, 2, \dots, 213$). The aggregate loss is given by $S = \sum_{i=1}^{213} X_i$, and $X_j$ corresponds to Apple Inc.

### 4.1. Long-term empirical experiment

In the first study, we analyze daily losses and factor data over a twenty-year period, from 1 January 2005 to 31 December 2024. The dataset contains 213 stocks that were continuously included in the S&P 500 during this period, each with 5033 daily losses. Here, $X_j$ represents the random loss of Apple Inc., one of the most heavily weighted constituents in the index.

Fig. 2 presents the empirical unconstrained and constrained risk bounds, along with $\mathrm{MES}_p(X_j, S)$, as functions of the prudence level $p$. We can conclude the following findings from the plot. First, the unconstrained risk bounds $M_j(p)$ and $m_j(p)$ indicate that the loss distribution of Apple is slightly left-skewed, since for symmetric distributions we would expect $M_j(p) = -m_j(p)$. Second, adding partial dependence information via the FFM structure does not improve the upper bound; i.e., $M_j(p) \approx M_j^f(p)$. For certain values of $p$, we have $M_j(p) < M_j^f(p)$. The estimated FFM introduces model-estimation error, so the estimated constrained bound may slightly exceed the estimated unconstrained bound. Essentially, the FFM is a multiple linear regression model built under specific assumptions. Third, the lower bound improves substantially when partial dependence is incorporated: $m_j^f(p)$ is much larger than $m_j(p)$, which assumes complete dependence uncertainty. Finally, the contribution of Apple's risk to systemic market risk, measured by $\mathrm{MES}_p(X_j, S)$, lies between $m_j(p)$ and $M_j(p)$. The constrained bounds $M_j^f(p)$ and $m_j^f(p)$ provide more realistic and practically useful ranges for MES.

The following subsections illustrate the relevance of these bounds in extreme market scenarios, specifically the 2007–2009 financial crisis and the COVID-19 pandemic.

### 4.2. The financial crisis

For this experiment, we use daily losses and factor data over a one-year period from 18 September 2007 to 17 September 2008. This period corresponds to the peak of the financial crisis, culminating in the bankruptcy and delisting of Lehman Brothers Holdings Inc. on 17 September 2008. The dataset contains 453 stocks that remained continuously in the S&P 500 during this period, each with 253 daily losses. Here, $X_j$ represents the random loss of Lehman Brothers.

Fig. 3 shows the empirical lower and upper bounds under complete uncertain and partially known dependence structures. In contrast to Apple's long-term case, the loss distribution of Lehman Brothers during the crisis is right-skewed, as indicated by $M_j(p)$ and $m_j(p)$. This indicates that its stock price exhibited a downward trend during the financial crisis. Moreover, the attainable MES during the crisis is ten times magnitude larger than in the long-term study, as seen from the scales of Figs. 2 and 3.

Crucially, the contribution of the individual risk induced by Lehman Brothers to market systemic risk, represented by the aggregated loss, nearly coincides with the constrained and unconstrained upper bounds, $M_j(p)$ and $M_j^f(p)$. This highlights the practical value of both the general upper bound and the constrained bound incorporating partial dependence information in Section 2. In worst-case scenarios, the collapse of a single institution, such as Lehman Brothers, can destabilize the entire financial system. It is evident that empirical $\mathrm{MES}_p(X_j, S)$ is quite close to both the unconstrained and constrained upper bounds. The constrained lower bound is substantially improved compared to the unconstrained one. Additionally, the degree of improvement for dependence-uncertainty spread through FFM is observed to be higher than the long-term case of Apple.

### 4.3. The pandemic downturn

The most recent major downturn was triggered by the pandemic, began with an outbreak of COVID-19 in December 2019. To further examine the practical usefulness of the bounds, we use daily losses and factors from 18 April 2019 to 23 March 2020. The start date corresponds to the listing of Zoom Communications Inc., while the end date marks the last day of the pandemic-induced downturn. The dataset includes 476 stocks that remained continuously in the S&P 500 (excluding Zoom) during this period, each with 234 daily losses. Unlike most firms, Zoom benefited from the pandemic due to increased demand for its services, making it an ideal case of an individual risk that contributes minimally—or even negatively—to systemic risk.

In Fig. 4, we present the empirical lower and upper bounds when the dependence information is completely unknown or partially known from the structure of FFM. The unconstrained bounds reveal that the loss distribution of Zoom during the pandemic is left-skewed, reflecting an increase in its stock price. Although the pandemic produced a narrower

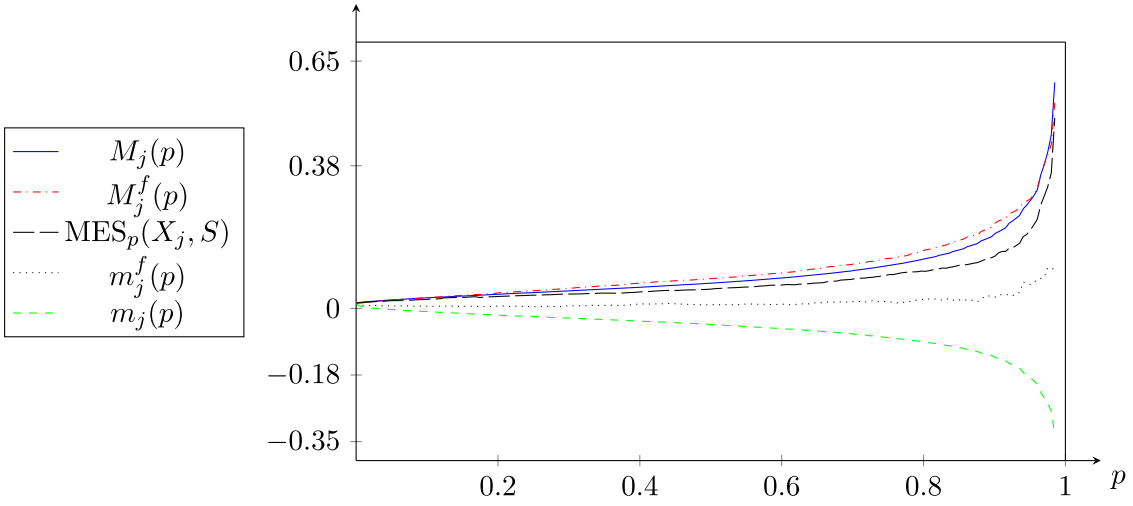



**Fig. 3.** Unconstrained and constrained risk bounds, together with the empirical values of $\mathrm{MES}_p(X_j, S)$, as functions of the prudence level $p$. $X_i$ denotes the daily loss of the $i$-th S&P 500 constituent that remained continuously in the index from 18 September 2007 to 17 September 2008 ($i = 1, 2, \dots, 453$). The aggregate loss is given by $S = \sum_{i=1}^{453} X_i$, and $X_j$ corresponds to Lehman Brothers Holdings Inc.

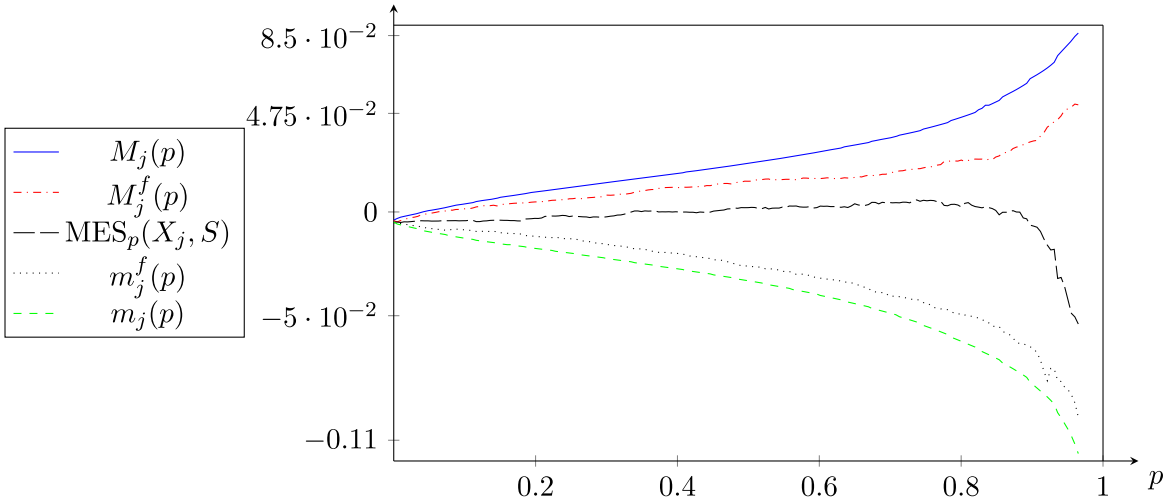



**Fig. 4.** Unconstrained and constrained risk bounds, together with the empirical values of $\mathrm{MES}_p(X_j, S)$, as functions of the prudence level $p$. $X_i$ denotes the daily loss of the $i$-th S&P 500 constituent that remained continuously in the index from 18 April 2019 to 23 March 2020 ($i = 1, 2, \dots, 476$). The aggregate loss is given by $S = \sum_{i=1}^{476} X_i$, and $X_j$ corresponds to Zoom Communications Inc.

attainable range of MES than the financial crisis, it resulted in a wider range than in the long-term case.

Interestingly, unlike in previous subsections, the unconstrained and constrained upper bounds no longer coincide, while the lower bounds tend to close with each other. The empirical $\mathrm{MES}_p(X_j, S)$ demonstrates that the contribution of the individual risk produced by Zoom to the systemic risk is negligible and, in some cases ($p > .8$), even reduces systemic risk. The lower bounds are thus attainable in practice, and the constrained upper bound provides a significant improvement over the unconstrained one. The empirical $\mathrm{MES}_p(X_j, S)$ decreases for large $p$ because Zoom's stock return tends to rise when the market declines during the pandemic downturn.

### 4.4. Systemic risk criticality index

To compare the systemic risk exposures of different companies, we define two SRCIs, for $p \in (0,\ 1)$ and $j \in \mathcal{N}$:

$$\mathrm{SRCI}_{j,p} = 1 - \frac{M_j(p) - \mathrm{MES}_p(X_j, S)}{M_j(p) - m_j(p)},$$

and

$$\mathrm{SRCI}^f_{j,p} = 1 - \frac{M^f_j(p) - \mathrm{MES}_p(X_j, S)}{M^f_j(p) - m^f_j(p)}.$$

SRCIs serve as practical tools for assessing the relative systemic position of firms within the market. Higher values indicate that a firm's observed MES lies closer to its worst-case dependence-consistent bound, suggesting greater systemic criticality and providing useful information for policymakers concerned with financial stability.

Table 1 reports the values of $\mathrm{SRCI}_{j,p}$ and $\mathrm{SRCI}^f_{j,p}$ for $p \in \{0.99, 0.993, 0.995\}$, covering a selection of companies that are currently or were previously included in the S&P 500 index. SRCIs for Apple, Coca Cola, Microsoft, and BNY Mellon are computed using daily loss data from 1 January 2005 to 31 December 2024, corresponding to the long-term analysis described in Section 4.1. The results for Lehman Brothers and Zoom, by contrast, are based on the periods from 18 September 2007 to 17 September 2008 and from 18 April 2019 to 23 March 2020, respectively, as discussed in Section 4.2.

**Table 1**
$\mathrm{SRCI}_{j,p}$ and $\mathrm{SRCI}^{f}_{j,p}$. The indices for Apple, Coca Cola, Microsoft and BNY Mellon are computed using daily loss data from 1 January 2005 to 31 December 2024, whereas those for Lehman Brothers and Zoom are based on the periods from 18 September 2007 to 17 September 2008 and from 18 April 2019 to 23 March 2020, respectively.

| Individual risk | Estimated $\mathrm{SRCI}_{j,p}$ | | | Estimated $\mathrm{SRCI}^{f}_{j,p}$ | | |
|---|---|---|---|---|---|---|
| | $p = .990$ | 0.993 | 0.995 | 0.990 | 0.993 | 0.995 |
| Apple | 0.832 | 0.839 | 0.823 | 0.737 | 0.768 | 0.737 |
| Lehman Brothers | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Zoom | 0.661 | 0.666 | 0.666 | 0.600 | 0.250 | 0.250 |
| Coca Cola | 0.859 | 0.839 | 0.823 | 0.857 | 0.846 | 0.840 |
| Microsoft | 0.838 | 0.864 | 0.865 | 0.647 | 0.718 | 0.751 |
| BNY Mellon | 0.921 | 0.932 | 0.927 | 0.965 | 0.970 | 0.951 |

As expected, Lehman Brothers during the global financial crisis exhibits the highest systemic risk importance across all three prudence levels, as measured by both $\mathrm{SRCI}_{j,p}$ and $\mathrm{SRCI}^{f}_{j,p}$, while Zoom during the COVID-19 pandemic shows the lowest systemic risk importance. Moreover, financial institutions such as BNY Mellon display higher systemic risk importance than firms in other industry sectors. For instance, $\mathrm{SRCI}_{j,0.995}$ equals 0.823 for Coca Cola but rises to 0.927 for BNY Mellon.

The empirical studies in this section confirm the theoretical properties of the unconstrained and constrained risk bounds developed in Section 2. For PPSs, financial regulators, and practitioners, these results may be useful for assessing systemic risk under worst-case dependence scenarios. In particular, such bounds can help inform risk-sensitive participation charges, reserve assessments, or capital-related decisions by highlighting the extent to which individual institutions may contribute to aggregate systemic distress. Moreover, PPSs could use the SRCIs to assess the relative systemic position of member firms within the market.

## 5. Conclusion

This paper studied the static marginal expected shortfall as a measure of institution-level contribution to aggregate systemic risk under dependence uncertainty. Motivated by prudential and policyholder-protection applications, we focused on the setting in which firms' individual risk distributions are known, while the dependence structure among firms is either completely unspecified or only partially specified.

We first derived sharp lower and upper bounds for $\mathrm{MES}_p(X_j, S)$ under fixed marginal distributions. These bounds characterize the full attainable range of systemic-risk contributions consistent with the observed marginal risks and therefore quantify the dependence-uncertainty spread of MES. We then showed how this spread can be reduced when partial dependence information is available through common-factor (or background-factor) structures, as well as through an alternative class of constraints based on a linear conditional-expectation relationship between individual and aggregate risks, connecting the analysis to CAPM-type and weighted-insurance-pricing frameworks.

The empirical analysis illustrated the practical relevance of the proposed bounds using financial loss data and the Fama–French five-factor model. The results show that partial dependence information can materially tighten the feasible MES range and that the position of an institution's empirical MES within this range varies meaningfully across firms and market regimes. In particular, firms under severe stress may lie close to their worst-case dependence-consistent upper bound, whereas firms that remain diversified from aggregate market distress may lie closer to the lower part of the attainable range.

Finally, we introduced the systemic risk criticality index, which evaluates how close a firm is to its own worst-case dependence-consistent systemic-risk contribution. This makes the index useful for comparing firms with different marginal risk profiles and for supporting applications in prudential supervision, systemic-risk monitoring, capital allocation, and risk-sensitive participation frameworks for policyholder protection schemes.

Overall, the paper argues that dependence uncertainty is a central feature of systemic-risk measurement. Accounting for this uncertainty through MES bounds and systemic risk criticality indices provides a transparent framework for assessing institution-specific systemic risk under both complete dependence uncertainty and partially specified dependence structures.

## CRediT authorship contribution statement

**Jinghui Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Conceptualization; **Edward Furman:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization; **X. Sheldon Lin:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

## Declaration of competing interest

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## Acknowledgments

The authors thank the three reviewers and the editor for their valuable comments and suggestions, which have helped improve the paper. Financial support from the project "New Order of Risk Management: Theory and Applications in the Era of Systemic Risk", provided through the Natural Sciences and Engineering Research Council of Canada (NSERC) grant ALLRP 580632 and the Mathematics of Information Technology and Complex Systems (Mitacs) grant IT33381, is gratefully acknowledged. X. Sheldon Lin acknowledges the financial support provided by the NSERC grant RGPIN-2023-04326. The authors also thank Yang Yang for his helpful remarks on the paper.

## Data availability

The CRSP data used in this study are available through Wharton Research Data Services subject to applicable licence restrictions. The Fama-French factors are available from the Kenneth R. French Data Library. Code and non-proprietary derived materials are available from the corresponding author on reasonable request.

## Appendix A. Proof of Theorem 1

Inequality chain (2.1) follows once we establish

$$\begin{aligned} X_j^a|S^a > \mathrm{VaR}_p(S^a) &\leq_{st} X_{j|Y}^a|S_{|Y}^a > \mathrm{VaR}_p(S_{|Y}^a) \\ &\leq_{st} X_j|S > \mathrm{VaR}_p(S) \\ &\leq_{st} X_{j|Y}^c|S_{|Y}^c > \mathrm{VaR}_p(S_{|Y}^c) \leq_{st} X_j^c|S^c > \mathrm{VaR}_p(S^c), \end{aligned} \tag{A.1}$$

where $\leq_{st}$ denotes the usual stochastic order or first order stochastic dominance. For random variables $X_1 \sim F_1$ and $X_2 \sim F_2$, $X_1 \leq_{st} X_2$ if and

only if $F_1(t) \geq F_2(t)$ for all $t \in \mathbb{R}$ (see, e.g., Theorem 2.2 of Bäuerle and Müller, 2006). Thus, proving (A.1) is equivalent to proving

$$F_{X_j^a|S^a>\mathrm{VaR}_p(S^a)}(t) \geq F_{X_{j|Y}^a|S_{|Y}^a>\mathrm{VaR}_p(S_{|Y}^a)}(t) \geq F_{X_j|S>\mathrm{VaR}_p(S)}(t)$$
$$\geq F_{X_{j|Y}^c|S_{|Y}^c>\mathrm{VaR}_p(S_{|Y}^c)}(t) \geq F_{X_j^c|S^c>\mathrm{VaR}_p(S^c)}(t),$$

for all $t$. Let $s_p = \mathrm{VaR}_p(S)$. First, we have

$$F_{X_j|S>s_p}(t) = \mathbb{P}(X_j \leq t|S > s_p) = \frac{\mathbb{P}(X_j \leq t, S > s_p)}{\mathbb{P}(S > s_p)}$$
$$= \frac{\mathbb{P}(X_j \leq t) - \mathbb{P}(X_j \leq t, S \leq s_p)}{\mathbb{P}(S > s_p)}. \tag{A.2}$$

Under the assumptions of factor models in (1.1),

$$\mathbb{P}(X_j \leq t, S \leq s_p) = \int \mathbb{P}\big(X_{j|y} \leq t, S_y \leq s_p\big) dH(y) \tag{A.3}$$

By the Fréchet-Hoeffding bounds for joint distribution functions (see, e.g., Nelsen, 2006),

$$\max(F_{j|y}(t) + F_{S_{|y}}(s_p) - 1, 0) \leq \mathbb{P}\big(X_{j|y} \leq t, S_y \leq s_p\big) \leq \min(F_{j|y}(t), F_{S_{|y}}(s_p)), \tag{A.4}$$

where the upper bound is obtained by $S_{|y}^c$ and $X_{j|y}^c$, and the lower is obtained by $S_{|y}^a$ and $X_{j|y}^a$. Therefore, by (A.2)-(A.4), for every $t$,

$$F_{X_{j|Y}^a|S_{|Y}^a>\mathrm{VaR}_p(S_{|Y}^a)}(t) = \frac{F_j(t) - \int \max(F_{j|y}(t) + F_{S_{|y}}(s_p) - 1, 0) dH(y)}{\mathbb{P}(S > s_p)}$$
$$\geq F_{X_j|S>s_p}(t)$$
$$\geq \frac{F_j(t) - \int \min(F_{j|y}(t), F_{S_{|y}}(s_p)) dH(y)}{\mathbb{P}(S > s_p)}$$
$$= F_{X_{j|Y}^c|S_{|Y}^c>\mathrm{VaR}_p(S_{|Y}^c)}(t).$$

Furthermore, $\min(F_{j|y}(t), F_{S_{|y}}(s_p)) \leq F_{j|y}(t)$ and $\min(F_{j|y}(t), F_{S_{|y}}(s_p)) \leq F_{S_{|y}}(s_p)$. Integrating both sides of these two inequalities, we have

$$\int \min(F_{j|y}(t), F_{S_{|y}}(s_p)) dH(y) \leq \min(F_j(t), F_S(s_p)).$$

Thus,

$$F_{X_{j|Y}^c|S_{|Y}^c>\mathrm{VaR}_p(S_{|Y}^c)}(t) \geq \frac{F_j(t) - \min(F_j(t), F_S(s_p))}{\mathbb{P}(S > s_p)} = F_{X_j^c|S^c>\mathrm{VaR}_p(S^c)}(t).$$

Similarly, we can prove that

$$\int \max(F_{j|y}(t) + F_{S_{|y}}(s_p) - 1, 0) dH(y) \geq \max(F_j(t) + F_S(s_p) - 1, 0),$$

and, then,

$$F_{X_{j|Y}^a|S_{|Y}^a>\mathrm{VaR}_p(S_{|Y}^a)}(t) \leq \frac{F_j(t) - \max(F_j(t) + F_S(s_p) - 1, 0)}{\mathbb{P}(S > s_p)}$$
$$= F_{X_j^a|S^a>\mathrm{VaR}_p(S^a)}(t).$$

## Appendix B. Proof of Proposition 1

For continuous RVs $X_i$,

$$\mathbb{E}[X_j^c|S^c > \mathrm{VaR}_p(S^c)] = \mathbb{E}[F_j^{-1}(U)|U > p] = \frac{\int_p^1 \mathrm{VaR}_q(X_j) dq}{1-p} = \mathrm{ES}_p(X_j),$$

where the components of $(X_1^c, X_2^c, \ldots, X_d^c)$ share a common tail event of probability $1-p$ (i.e., $X_i^c \mathbb{1}_{X_i^c > \mathrm{VaR}_p(X_i^c)} = F_i^{-1}(U \mathbb{1}_{U>p})$), and

$$\mathbb{E}[X_j^a|S^a > \mathrm{VaR}_p(S^a)] = \mathbb{E}[F_j^{-1}(U)|U < 1-p] = \frac{\int_0^{1-p} \mathrm{VaR}_q(X_j) dq}{1-p}$$
$$= \mathrm{LES}_{1-p}(X_j).$$

## Appendix C. Proof of Proposition 3

Under the bivariate normal setting, it is well-known that $\mathbb{E}[X|S] = \alpha + \beta S$, where $\alpha = \mu_X - \beta\mu_S$ and $\beta = \rho\frac{\sigma_X}{\sigma_S}$; see, e.g., Furman and Zitikis (2017b). An application of Theorem 2 then yields

$$\mathrm{MES}_p(X, S) = \mu_X + \rho\frac{\sigma_X}{\sigma_S}\big(\mathrm{ES}_p(S) - \mu_S\big),$$

Since $\rho \in [-1, 1]$ and all other terms are constants, the corresponding risk bounds follow as shown in Eq. (3.1).

## Appendix D. Proof of Theorem 3

We first prove the case of $d$ non-negative risks. Since $\sum_{i=1}^d \mathbb{E}[X_i|S = 0] = \mathbb{E}[S|S = 0] = 0$ and $\mathbb{E}[X_i|S = 0] \geq 0$ for all $i$, it follows that $\alpha = 0$ if $\mathbb{E}[X_j|S] = \alpha + \beta S$. In addition,

$$\beta = \frac{\mu_j}{\sum_{i=1}^d \mu_i},$$

because $\mathbb{E}[\mathbb{E}[X_j|S]] = \mathbb{E}[X_j] = \beta\mathbb{E}[S]$. Since $\Pi_w(X_j, S) = \mathrm{MES}_p(X_j, S)$ for $w(s) = \mathbb{1}_{s>t}$, Theorem 2 applies. Then,

$$\mathrm{MES}_p(X_j, S) = \mu_j + \frac{\mu_j}{\sum_{i=1}^d \mu_i}\left(\mathrm{ES}_p(S) - \sum_{i=1}^d \mu_i\right) = \frac{\mu_j}{\sum_{i=1}^d \mu_i}\mathrm{ES}_p(S). \tag{D.1}$$

In this case, solving for risk bounds on $\mathrm{MES}_p(X_j, S)$ reduces to solving for bounds on $\mathrm{ES}_p(S)$. By Propositions 3.2 and 3.5 of Bernard et al. (2017b),

$$\sum_{i=1}^d \mu_i = \mathrm{ES}_p\left(\sum_{i=1}^d \mu_i\right) \leq \mathrm{ES}_p(S) \leq \mathrm{ES}_p\left(\sum_{i=1}^d F_i^{-1}(U)\right) = \sum_{i=1}^d \mathrm{ES}_p(X_i).$$

The last equation holds because ES is comonotonic additive and objective. The risk bounds on $\mathrm{MES}_p(X_j, S)$ are thus obtained by Eq. (D.1) and $\beta \geq 0$.

The proof for non-positive risks is similar, with the crucial observation that $\beta \geq 0$ in both cases.

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